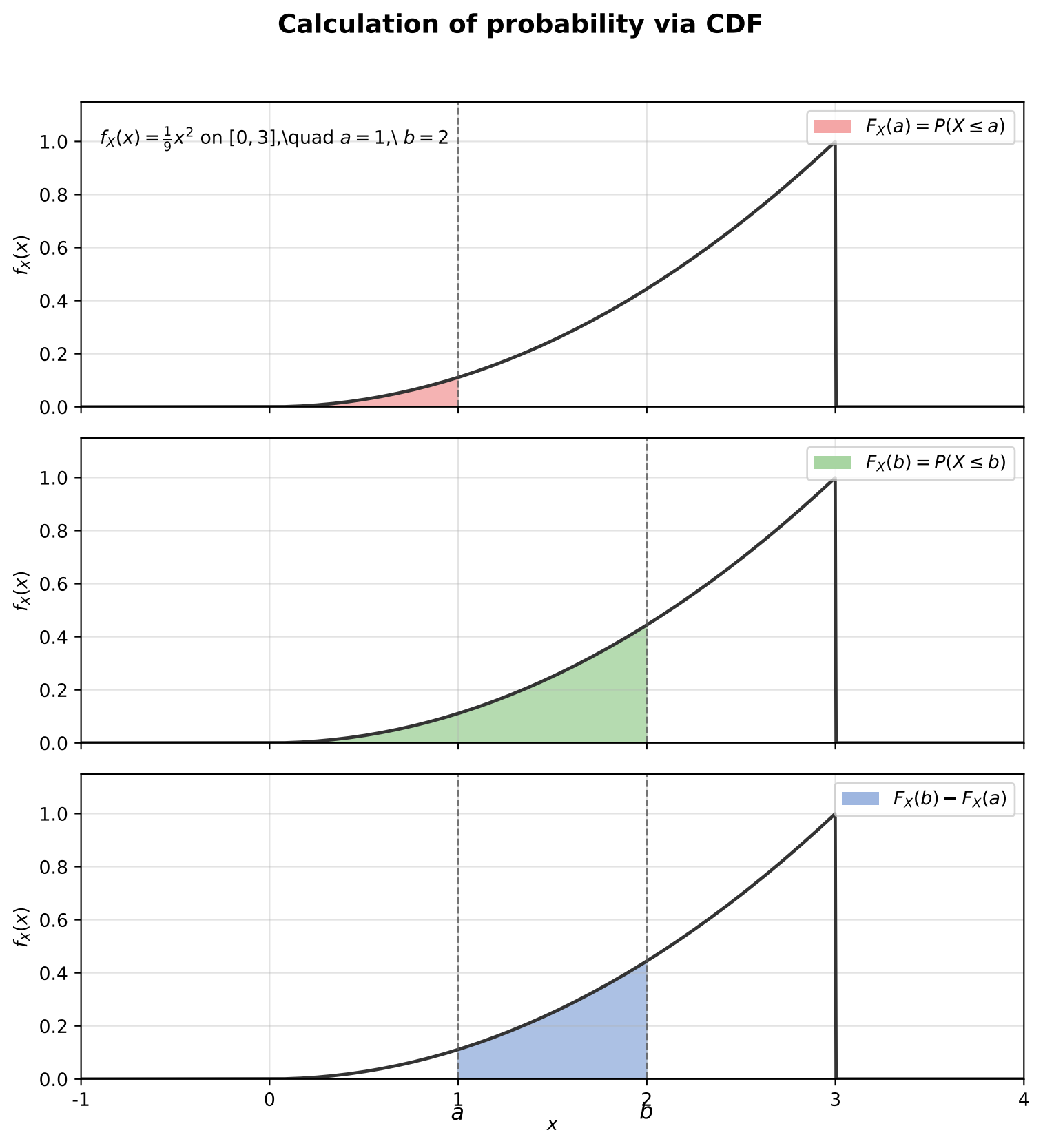
Calculation of probability via CDF
$f_X(x) = \frac{1}{9}x^2$ on $[0, 3]$,\quad $a = 1$,\ $b = 2$
$F_X(a) = P(X \le a)$
$F_X(b) = P(X \le b)$
$F_X(b) - F_X(a)$
$f_X(x)$
x
a
b
0.0
0.2
0.4
0.6
0.8
1.0
-1
0
1
2
3
4